

Esra Onal

Brain and Psychological Sciences
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Research Statement

Cognitive scientist with an interdisciplinary background in linguistics, computer science, and cognitive science, studying anaphora resolution to test whether transformer models process language in human-like ways.

Education

Indiana University of Bloomington

Jan 2021 – Present

Joint Ph.D., Computational Linguistics & Cognitive Science

Bloomington, IN, USA

- Main Advisor: Francis M. Tyers (Computational Linguistics)
- Co-advisor: Joshua Brown (Psychological and Brain Sciences)

Boğaziçi University

August 2017 – August 2021

M.A., Cognitive Science

Istanbul, Türkiye

- Advisor: Arzucan Özgür Türkmen (Computer Engineering)
- Co-advisor: Balkız Öztürk (Linguistics)
- Thesis: "MorphLaz: A finite-state morphological analyzer for Laz"

Boğaziçi University

August 2012 – August 2017

B.A., Foreign Language Education with a minor in Linguistics

Istanbul, Türkiye

Experience

Associate Instructor

January 2022 - Present

Indiana University, Linguistics Department & Cognitive Science Program

Bloomington, IN, USA

- COGS 320 - Computation in Cognitive Science, Spring 2026
- COGS 350 - Math and Logic in Cognitive Science, Fall 2025
- COGS 240 - Philosophical foundations of Cognitive Science, Fall 2025
- C104 - Language and Cognition, Spring 2023/2024
- LING103 - Introduction to the study of language, Spring 2022 (Instructor of Record)

Data Analyst & Research Assistant

Summer 2021 & 2022

Indiana University, Digital Arts & Humanities

Bloomington, IN, USA

- Bibliographic reference extraction and citation network analysis

Research Assistant

January 2021 – January 2022

Indiana University, Linguistics Department

Bloomington, IN, USA

- Lab: Language Technologies for Marginal Languages

Visiting Scholar

January 2019 – May 2019

Indiana University, Computational Linguistics

Bloomington, IN, USA

- Project: Developing a morphological analyser for the endangered Laz Language

Manuscripts

Onal, Esra and Kübler, Sandra. *Syntactic Prominence and Pragmatic Bias in Turkish Subject Anaphora: Humans and Large Language Models*. Manuscript submitted.

Onal, Esra. *Measuring Types of Semantic and Pragmatic Inference in Large Language Models Using Embedding-Based Similarity*. Manuscript in preparation.

Peer-Reviewed Conference Papers

Mirzakhlov, J., Babu, A. S., Ataman, D., Kariev, S., Tyers, F. M., Abduraufov, O., Hajili, M., Ivanova, S., Khaytbaev, A., Laverghetta, A., Moydinboyev, B., **Onal, Esra**, Pulatova, S., Wahab, A., Firat, O., & Chellappan, S. (2021). *A Large-Scale Study of Machine Translation in the Turkic Languages*. Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP).

Onal, Esra and Tyers, F. M. (2019). *Building a Morphological Analyser for Laz*. Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP), 870–878.

Conference Talks and Posters

Onal, Esra. (2026). *Investigating the Cognitive Plausibility and Linguistic Interpretability of Large Language Models through Anaphora Resolution*. Presentation at the 39th Annual Conference on Human Sentence Processing (HSP), MIT, Cambridge, MA, USA.

Onal, Esra. (2025). *Interpretation and Antecedent Biases of Null and Overt Pronouns in Turkish*. Oral presentation at the 38th Annual Conference on Human Sentence Processing (HSP), University of Maryland, USA.

Onal, Esra and Tyers, F. M. (2025). *Rethinking the Distinction Between Depictives and Adverbials in Turkish*. Talk at the Workshop on Turkic and Languages in Contact with Turkic (Tu+10), University of Southern California, USA.

Onal, Esra and Tyers, F. M. (2019). *A Finite-State Morphological Analyser for Pazar Laz*. Oral presentation at the International Conference on Language and Modern Technologies V (LMT), Tbilisi State University, Georgia.

Onal, Esra. (2017). *Action Research: Motivating Silent Students in Language Classrooms*. Poster presented at Teachers Research Istanbul (IATEFL), Bahçeşehir University, Turkey.

Reviewing Service

Journal, Conference and Workshop Peer Reviewer:

- SyntaxFest 2021 — *Universal Dependencies for Old Turkish* (workshop)
- LREC 2026 — *AmbiCoRefVis: A Tool for Visualizing Coreferential Ambiguity*
- LCIU Working Papers 2025 — *Exploration of Feminist Identities Through Protest Language in Türkiye*

Awards & Honors

Conference Travel Award <i>Indiana University of Bloomington</i>	2025
Visiting Graduate Fellowship <i>Boğaziçi University</i> • Development and curation of a Laz speech corpus for Mozilla Common Voice to support open, low-resource speech technologies	2023
IU Graduate Fellowship <i>Indiana University of Bloomington</i>	2021
Research Conference Travel Award (2×) <i>UYGAR Language Center, Boğaziçi University</i>	2019
Visiting Scholar Fellowship <i>Indiana University of Bloomington</i> • Development of a finite-state morphological analyzer for Laz	2019
Graduate Scholarship <i>Scientific and Technological Research Council of Turkey (TÜBİTAK)</i>	2018–2020
Honors Certificate <i>Faculty of Education, Boğaziçi University</i>	2017

Interests/Activities

- Member of *Turkic Interlingua* (TIL)
- Member of *Tehlike Altındaki Diller* (TADNET) in Turkey (an endangered languages network based in Turkey)
- Member of the *Inclusive Technology for Marginalised Languages* (ITML) lab in Department of Linguistics at Indiana University
- 2022-2023 Student representative of *Graduate and Professional Student Government* (GPSG) for Department of Linguistics at Indiana University of Bloomington in Indiana, US
- 2022-2024 President and Vice President of *Linguistics Club at Indiana University* (LCIU)
- 2015-2019 Volunteer and project coordinator for *Aid Foundation for Elementary Schools* (ILKYAR) and *Education and Research Community* at Boğaziçi University, Turkey
- 2017-2018 President of the *Aid Foundation for Elementary Schools* (ILKYAR) community at Boğaziçi University

References

- Prof. Sandra Kübler**
Department of Linguistics, Indiana University of Bloomington, Indiana IN, USA
E-mail: skuebler@iu.edu
- Assoc. Prof. Francis M. Tyers**
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